Port City International University

REPORT

Report Name : Study on Tools of Maintenance

Report No : 01

Course Code : TEX 471

Course Title : Maintenance of Fabric Manufacturing Machinery LAB

Submitted To	Submitted By
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Name of Experiment: Study on Tools of Maintenance

1. Introduction:

Maintenance is a very important activity in any industry to keep machines, equipment, and tools in good working condition. It helps to increase the efficiency, safety, and lifespan of machines. There are different types of maintenance such as preventive maintenance, corrective maintenance, and breakdown maintenance. Preventive maintenance is done regularly to avoid unexpected failures, while corrective maintenance is performed to fix faults.For proper maintenance work, different types of tools are required. Each tool has its own specific function and is used for a particular purpose. Using the correct tool not only makes the work easier but also ensures safety and accuracy. Lack of proper tools or incorrect use may lead to accidents, machine damage, and production loss. Therefore, it is essential to have proper knowledge about maintenance tools, their types, and their functions. This experiment helps us to understand different categories of maintenance tools and their practical uses.

2. Objectives:

- To know about maintenance
- To know about maintenance tools
- To know about different types of tools
- To know about the functions of different tools

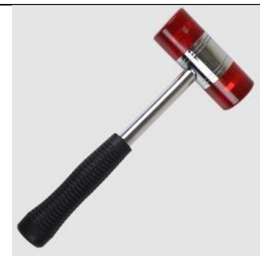
3. Function of Maintenance Tools:

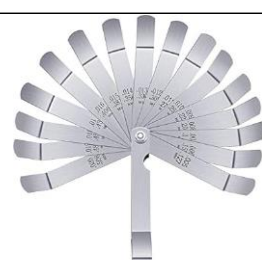
SL	Tools Type	Examples	Figure	Function
1	Safety Tools	Gloves, Helmet, Goggles		To ensure safety of workers during maintenance work
2	Cutting Tools	Knife, Scissors, Wire cutter		To cut wires, materials, or fabrics

3	Holding Tools	Pliers, Clamps		To hold objects firmly during work
4	Striking Tools	Hammer		To hit or shape objects
5	Tightening Tools	Spanner, Screwdriver, Adjustable wrench		To tighten or loosen nuts, bolts, and screws
6	Measuring Tools	Measuring tape, Scale		To measure length, size, or distance
7	Testing Tools	Multimeter		To test electrical values like voltage and current
8	Lubricating Tools	Oil can		To apply oil for smooth machine operation

9	Drilling Tools	Hand drill		To make holes in materials
10	Precision Tools	Feeler gauge		To measure small gaps or clearances accurately
11	Spanner Tools	Open-end spanner		Tightening and loosening nuts and bolts
12	Ring Spanner Tools	Double ring spanner		Provides better grip on bolts
13	Adjustable Wrench Tools	Monkey wrench		Fits different sizes of nuts and bolts
14	Socket Wrench Tools	Ratchet socket set		Fast tightening and loosening

15	Screwdriver Tools	Flat head screwdriver		Tightening or removing screws
16	Phillips Screwdriver Tools	Star screwdriver		Used for cross-head screws
17	Allen Key Tools	Hex key set		Tightening hexagonal socket screws
18	Pliers Tools	Combination pliers		Holding, bending, and cutting wires
19	Nose Pliers Tools	Long nose pliers		Working in narrow spaces
20	Cutting Pliers Tools	Side cutter		Cutting wires and small metals

21	Rubber Mallet Tools	Soft mallet		Prevents surface damage during striking
22	Chisel Tools	Cold chisel		Cutting or shaping metal
23	File Tools	Flat file		Smoothing rough metal surfaces
24	Hacksaw Tools	Hand hacksaw		Cutting metal rods and pipes
25	Pipe Wrench Tools	Stillson wrench		Holding and rotating pipes
26	Torque Wrench Tools	Click torque wrench		Applying exact tightening force

27	Grease Gun Tools	Manual grease gun		Lubricating machine parts
28	Oil Can Tools	Hand oil can		Applying lubricating oil
29	Bearing Puller tools	Gear puller		Removing bearings and gears
30	Multimeter Tools	Digital multimeter		Measuring voltage, current, and resistance
31	Clamp Meter Tools	AC clamp meter		Measuring electrical current safely
32	Feeler Gauge Tools	Steel feeler gauge		Measuring small gaps and clearances

33	Vernier Caliper tools	Digital vernier caliper		Accurate measurement of dimensions
34	Micrometer Tools	Outside micrometer		Measuring thickness and diameter precisely
35	Spirit Level Tools	Bubble level		Checking horizontal and vertical alignment
36	Drill Machine Tools	Electric hand drill		Drilling holes in materials
37	Grinding Machine Tools	Angle grinder		Cutting and smoothing metal surfaces
38	Welding Machine Tools	Arc welding machine		Joining metal parts together

39	Air Compressor Tools	Portable compressor		Supplying compressed air for tools and cleaning
40	Chain Wrench Tools	Chain pipe wrench		Holding and turning round objects and pipes
41	Crimping Tools	Wire crimping pliers		Joining electrical terminals and wires
42	Wire Stripper Tools	Automatic wire stripper		Removing insulation from wires
43	Soldering Iron Tools	Electric soldering iron		Joining electrical components with solder
44	Heat Gun Tools	Electric heat gun		Heating, drying, and shrinking materials

45	Hydraulic Jack Tools	Bottle jack		Lifting heavy machinery and equipment
46	Bench Vice Tools	Cast iron vice		Holding workpieces firmly during maintenance
47	Tap and Die Set Tools	Hand tap set		Creating or repairing screw threads
48	Inspection Mirror Tools	Telescopic inspection mirror		Viewing hidden machine parts
49	Flashlight Tools	LED torch light		Illuminating dark work areas
50	Vacuum Cleaning Tools	Industrial vacuum cleaner		Cleaning dust and waste from machines

4. Remarks:

- Proper selection of tools is necessary for efficient work
- Safety tools must always be used during maintenance
- Regular checking and maintenance of tools are important

5. Conclusion:

From this experiment, we gained knowledge about different types of maintenance tools and their functions. We learned that using the right tools in the right way is essential for safe and efficient maintenance work. Proper understanding of tools helps to improve work quality and reduce the risk of accidents.